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MILITARY SPECIFICATION

TELEPHONE CENTRAL OFFICE GROUP, MANUAL AN/GTA-6( )

1. SCOPE

1.1 This specification covers one type of telephone central group designated Telephone Central Office Group, Manual AN/GTA-6( ) hereinafter referred to as the "telephone central group".

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-S-801	Sulphuric Acid, Electrolyte (For Storage Batteries)
GG-H-941	Hydrometers; Syringe, Battery

Military

MIL-M-14	Molding Plastics and Molded Plastic Parts; Thermosetting
MIL-J-641	Jacks, Telephone, General Specification For
MIL-P-642	Plugs, Telephone
MIL-E-4158	Electronic Equipment; Ground: General Requirements For
MIL-M-4552	Multimeter AN/PSM-6
MIL-T-4807	Tests, Vibration and Shock, Ground Electronic Equipment; General Requirements For
MIL-N-7513	Nomenclature Assignment and Nameplate Approval; Contractors Method for Obtaining
MIL-C-10392	Cables, Special Purpose, Electrical (Miniature)
MIL-G-11994	Generator, Ringing, Hand G-42( )/PT

FSC 5805

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MIL-H-13253	Handsets [Handsets TS-90( ), TS-15( ), H-22( )/U, H-23( )/U, H-90/U, H-94( )/U, H-100( )/U, H-111( )/U, H-121( )/U, H-122( )/U and H-123( )/U]
MIL-C-13294	Cable, Telephone Electrical (Infantry Field Wire, Twisted Pair, Wire WD-1/TT and WD-14/TT)
MIL-T-13626	Telephone Set TA-1( )/PT
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-T-14358	Telephone TA-312( )/PT
MIL-B-15072	Batteries, Storage, Lead-Acid, Portable (Except for Aircraft and Automotive Vehicles)
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-R-26474	Reliability Requirements for Production Ground Electronic Equipment
MIL-M-26512	Maintainability Requirements for Aerospace Systems and Equipment
MIL-H-26827	Handset Headset H-108( )/4 and Cord CX-1245( )/W (7 ft. 6 in.)
MIL-H-27894	Human Engineering Requirements for Aerospace Systems and Equipment

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports, Preparation of

DRAWINGS

U.S. Air Force

50A54009	Schematic Diagram, C.B. Line Terminal PBX Switchboard
54A54386	Telephone Central Office Group, Manual AN/GTA-6( )
54A54387	Schematic Diagram, Magneto and Night Alarm, PBX Switchboard

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54A54393	Wiring Diagram, Protector Alarm Circuit
50B54616	Schematic Diagram, Moisture Proof Telephone, Telephone Central Group
54B54005	Adapter Bar, Operator's 10 Line Unit
54B54006	Bar, Operator's 10 Line Unit
54B54007	Stud, Bar, Operator's 10 Line Unit
54B54388	Schematic Diagram, Trunk Circuit, PBX Switchboard
54B54390	Key Shelf Arrangement, PBX Switchboard
54B54391	Schematic Diagram, Operator's Tel. Ckt., PBX Switchboard
54B54392	Wiring Diagram, Wire Chief's Test Panel
54B54394	Wiring Diagram, Fuse Alarm Ckt., D.C. Only
50C54006	Equipment Arrangement, PBX Switchboard, Telephone Central Group
51C55404	Wiring Diagram, Universal Cord Circuit
54C54389	Jack Panel Arrangement, Relay Rack #1
54C54395	Face Equipment, PBX Switchboard
54C54396	Terminal Arrangement, PBX Switchboard
54C54397	Instruction Label, Front Cover, Battery
48D55619	Communication Equipment, Volume Limiter
51D55061	Cable Diagram
51D55340	Schematic Diagram, Light Weight Power Panel
54D54398	Outline-Dimensions, Wire Terminal Box
54D54399	Outline-Dimensions, Power Panel
54D54400	Wiring Diagram, 5 Wire Ring Down Trunk
54D54401	Outline Dimensions, Volume Limiter Box

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

The Clearing House for Federal Scientific and Technical Services (CFSTS)

PB-161894	RADC Reliability Notebook RADC-TR-58-111

(Copies of CFSTS documents may be obtained from The Clearing House for Federal Scientific and Technical Services (CFSTS) U.S. Dept. of Commerce, Wash. 25, D.C. or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The telephone central group shall consist of the following components:

Item	Quantity	Description	Requirement
1	10 each	Telephone Set TA-312( )/PT	3.7.1
2	20 each	Telephone TA-236/FT	3.7.2
3	2 each	Telephone TA-168( )/GT	3.7.3
4	40 each	Handset H-67( )/GT	3.7.4
5	40 each	Handset-Headset H-108( )/U	3.7.5
6	40 each	Cord CX-1245( )/U (7 ft. 6 in.)	3.7.6
7	1 each	Manual Telephone Switchboard SB-202( )/GTA-4	3.7.7
8	1 each	Main Distributing Frame, Telephone TA-270( )/GTA-6	3.7.8.1
9	1 each	Main Distributing Frame, Telephone TA-285( )/GTA-6	3.7.8.2
10	1 each	Relay Rack, Telephone TA-268( )/GTA-6	3.7.9
11	1 each	Power Supply PP-706( )/GTA-21	3.7.10
12	1 each	Control, Telephone Line C-1421( )/GTA-6	3.7.11
13	1 each	Multimeter AN/PSM-6	3.7.12
14	4 each	Terminal Box J-618( )/GT	3.7.13
15	25 each	Telephone Circuit, Operator's TA-277( )/GTA-6	3.7.14
16	3 each	Ground, Rod GP-26( )	3.7.15
17	3 each	Clamp, Ground Rod	3.7.16
18	12 each	Adapter Bracket	3.7.17

3.2.1 Cables, cords and wire. The following cables, cords and wire shall be provided:

Item	Quantity	Description	Requirement
1	20 each	Switchboard Cord CX-1220( )/U (1 ft. 6 in.)	3.7.18
2	10 each	Switchboard Cord CX-1220( )/U (2 ft. 0 in.)	3.7.18
3	10 each	Switchboard Cord CX-1220( )/U (4 ft. 0 in.)	3.7.18
4	15 each	Cord Assembly, Electrical CX-2763( )/U (approximately 12 inches)	3.7.19

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5	3 each	Electrical Lead CX-1841( )/U (10 ft. 0 in.)	3.7.20
6	2 each	Electrical Lead CX-1841( )/U (25 ft. 0 in.)	3.7.20
7	2 miles	Wire WD-1( )/TT	3.7.21
8	2000 ft.	Wire, Cross Connecting	3.7.22
9	1 each	Cable Assembly, Power, Electrical CX-2795( )/U (20 ft. 0 in.)	3.7.23
10	2 each	Cable Assembly, Telephone CX-2921( )/U (25 ft. 0 in.)	3.7.24
11	3 each	Cable Assembly, Telephone CX-2921( )/U (10 ft. 0 in.)	3.7.24
12	6 each	Cable Assembly, Telephone CX-2873( )/U (10 ft. 0 in.)	3.7.25
13	3 each	Cable Assembly, Telephone CX-2873( )/U (30 ft. 0 in.)	3.7.25
14	3 each	Cable Assembly, Telephone CX-2873( )/U (50 ft. 0 in.)	3.7.25
15	8 each	Cable Assembly, Telephone CX-2922( )/U (10 ft. 0 in.)	3.7.26
16	25 each	Cable Assembly, Telephone CX-2922( )/U (50 ft. 0 in.)	3.7.26
17	2 miles	Wire, Insulated Common Battery Hook-Up	3.7.27

3.3 General specification. The requirements of MIL-E-4158 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3.1 Service conditions. Servcie conditions shall conform to MIL-E-4158 except as specified below.

3.3.1.1 Ambient temperature

a. Operating minus 29°C (-20°F) to plus 60°C (140°F)

b. Nonoperating minus 54°C (-65°F) to plus 71°C (160°F)

3.3.1.2 Other service conditions. Other service conditions shall include:

a. Fungus

b. Salt fog

c. Sand and dust

3.3.1.3 Mechanical. The following service conditions shall apply:

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a. Shock

b. Bounce

c. Vibration

3.3.2 Interference. The telephone central group shall conform to the limits specified for Class G_p equipment in accordance with MIL-STD-826.

3.3.3 Structural welding. The requirement for qualification of welders shall not apply.

3.3.4 Relays. Line relays connected to ring-down trunks or to multiple line terminals shall operate on 75 volts AC, 20 CPS through a resistance of not less than 3000 ohms.

3.3.5 Aluminum. The aluminum used shall be of the proper alloy and hardness to provide necessary strength and rigidity.

3.3.6 Connectors, electrical. Waterproof connectors shall be used in the construction of the telephone central group.

3.3.7 Identification markings and labels. The reference number of cables shall be identical with the numbers assigned in accordance with Drawing 51D55061.

3.3.8 Wire hook-up. Hook-up wire used in cable forms of telephone equipment shall be type LW, No. 22 AWG, solid in accordance with MIL-W-16878.

3.3.9 Human engineering. In addition to the requirements of MIL-E-4158 the contractor shall apply human engineering design principles which shall maximize the military effectiveness of man-equipment combinations, and minimize unnecessary demands upon human skill, training, and manpower. This requirement shall be accomplished by the utilization of the human engineering requirements specified in MIL-H-27894.

3.3.10 Maintainability. The telephone central group shall have a mean time to repair (MTTR) of no greater than 45 minutes (see 6.2.e).

3.3.11 Reliability. The required degree of reliability in the design and construction of the equipment shall be accomplished in accordance with MIL-R-26474 (see 6.2.e).

3.4 Nomenclature. When required by the contract the nomenclature shall be obtained in accordance with the requirements of MIL-N-7513. The parentheses appearing in the nomenclature may be dropped or be replaced by a letter to designate the particular design (see 6.2.g).

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3.5 Design and construction. The telephone central group shall serve as a communication and monitoring network for fixed or mobile radar sets. By means of this network, it shall be possible to accomplish the following:

a. Selective intercommunication among the operators of the radar set.

b. Communication between the radar shelters and information centers, or other radar sets and other outside points, including air to ground, by telephone lines or by control of radio link.

3.5.1 The general layout of cabling and schematic design shall be in accordance with the drawings on the associated drawing list, Drawing 54A54386.

3.5.2 Power source. All telephone circuits shall be designed to operate at 24 plus or minus 4 volts DC.

3.5.3 Earth grounds. There shall be no earth ground connections to any of the equipment specified herein with the exception of the main frame.

3.5.4 Telephone circuit (radio or wire). Telephone circuit (radio or wire) shall be used at all operating locations requiring communication facilities over either radio channels or telephone lines. The telephone circuit shall consist of a terminating network for matching a Handset TS-90( ), MIL-H-13253, to a balanced 600-ohm telephone line or a two-wire line to a radio control channel. The transmitter current for the handset shall be brought in through two 100-ohm windings of a control relay in series with the microphone. The control relay shall operate when the push-to-talk switch is closed, connecting indirect negative battery to a control wire.

3.5.5 Radio control circuit. There shall be a radio control circuit which shall function when any telephone circuit (radio or wire) is connected with a radio channel. It shall include a control relay connected to the control wire which shall operate when the control wire is connected to indirect negative battery by the operation of the relay in the telephone circuit. The operation of the control relay shall transfer the output of the terminating network from a pair of conductors going to the radio receiver to a pair of conductors going to the radio transmitter.' The operation of the control relay shall also close two contacts which shall be used to control the output of the radio transmitter.

3.5.6 Air transportability. Components of the telephone central group shall be air transportable in type C-119 aircraft. Individual

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components shall weigh not more than 400 pounds prior to packaging, and the floor loading shall not exceed 200 pounds per square foot.

3.5.7 Cabinets. The cabinets of the telephone main frame and the relay rack shall be designed to protect the equipment from damage when subjected to the sand and dust test (see 4.6.4) and shall be capable of meeting the following requirements.

3.5.7.1 Openings. Openings required during operation shall be provided with quick detachable fasteners.

3.5.7.2 Mounting. Each cabinet shall be capable of being fastened to the floor of a shelter or trailer. For private branch exchange (PBX) switchboards and main frame sections, the floor-securing device shall not extend beyond the plane of the sides of the units so that the sides of adjacent units shall be in contact with each other.

3.5.8 Terminal boards. Plastic used for terminal boards shall be in accordance with MIL-M-14, Type CFG.

3.6 Performance.

3.6.1 Insulation resistance and dielectric strength. The total insulation resistance of all circuit loops shall not be less than 10 megohms.

3.6.2 Crosstalk. The vacuum tube voltmeter reading of the disturbed circuit shall not exceed 0.001 volt RMS with a potential of not less than 2 volts RMS on the disturbing circuit.

3.7 Details of components.

3.7.1 Telephone Set TA-312( )/PT. Telephone Set TA-312( )/PT shall be in accordance with MIL-T-14358.

3.7.2 Telephone TA-236/FT. Telephone TA-236/FT shall be a common battery deck set with a self contained ringer type incoming signal device contained in the base, with a hook and switch type outgoing signal device and shall be used for local telephone stations. One two-wire connecting block shall be furnished with each telephone. The cord from the handset to the telephone set shall be coiled.

3.7.3 Telephone TA-168( )/GT. The telephone TA-168( )/GT shall be a sound powered magneto type telephone in accordance with the standard sample and inclosed in a metal cabinet with parts treated to resist the action of moisture. Telephone TA-168( )/GT shall be wired in accordance with Drawing 50B54616 and shall include the following:

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a. 1 each Handset TA-1

b. 1 each handset cord

c. 1 each Generator, Ringing, Hand G-42( )/PT

d. 1 each cabinet

3.7.3.1 Handset TA-1( ). Handset TA-1( ) shall consist of a sound-powered microphone and receiver unit in a suitable mounting in accordance with MIL-T-13626.

3.7.3.2 Handset cord. The handset cord, 36 inches in length when extended, shall be a three-conductor, tinsel-coiled cord.

3.7.3.3 Generator, Ringing, Hand G-42( )/PT. Generator, Ringing, Hand G-42( )/PT shall be in accordance with MIL-G-11994.

3.7.3.4 Cabinet. The cabinet shall include a gasket door and no other openings except two holes in the bottom for entrance of line wires. These holes shall be provided with grommets which shall effectively close the openings against moisture when the wires are in place. The cabinet shall have a latch which cannot be obstructed by ice or snow and shall be capable of being operated by men wearing arctic type gloves. The cabinet shall be capable of being attached to a pole or an antenna mount. The equipment in the cabinet, with the exception of the handset, shall be protected by a plastic inner panel. The bottom of the cabinet shall slope towards the door. Drain holes shall be provided through the bottom of the plastic inner panel.

3.7.4 Handset H-67( )/GT. Handset H-67( )/GT shall have a coiled cord terminated in Plug PJ-068, MIL-P-642. The cord shall extend to 3-1/2 feet. The handset received impedance shall be 250 ohms and the transmitter resistance shall be 25 ohms. The handset shall contain a push-to-talk switch in the transmitter circuit. The handle shall be black phenolic with overall dimensions of 9-1/8 inches long by 2-3/4 inches wide by 3-15/16 inches deep.

3.7.5 Handset-Headset H-108( )/U. The handset-headset shall be in accordance with MIL-H-26827.

3.7.6 Cord CX-1245( )/U (7 ft. 6 in.). The cord shall be in accordance with MIL-H-26827.

3.7.7 Manual Telephone Switchboard SB-202( )/GTA-4. The switchboard shall consist of one section with common battery and magneto

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circuits. All openings shall be provided with gasket doors, secured with approved quick release fasteners. A cover shall be provided to protect the key shelf and fact equipment during shipment or storage. The section shall contain the following circuits:

	No. of lines	Circuit
a.	20	Magneto lines in accordance with Drawing 54A54387
b.	40	Common battery lines in accordance with Drawing 50A54009
c.	10	Universal trunks in accordance with Drawing 54B54388
d.	10	Universal cord circuits in accordance with Drawing 51C55404
e.	1	Operator's telephone circuit (wire only) in accordance with Drawing 54B54391
f.	1	Operator's dialing circuit in accordance with Drawing 54B54388
g.	2	Night alarm circuit in accordance with Drawing 54A54387
h.	1	Emergency ringing circuit in accordance with Drawing 54B54391
i.	1	Circuit for connecting switchboard sections in accordance with Drawing 54B54391

3.7.7.1 Connectors. Connectors shall be provided for three 24-pair and three five-pair cables. Two ground studs shall be provided for the ground wires.

3.7.7.2 Terminal boards. Terminal boards shall be provided for 70 two-wire circuits.

3.7.7.3 Binding posts. Twenty pairs of binding posts shall be provided for connecting common battery extensions directly to the switchboard. The connections of the five-pair cables shall be in accordance with Drawing 54B54391.

3.7.7.4 Switchboard arrangement. The arrangement of the switchboard shall be as follows:

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a. Jack panel arrangement shall be in accordance with Drawing 54C54395.

b. Key shelf arrangement shall be in accordance with Drawing 54B54390.

c. Equipment arrangement shall be in accordance with Drawing 50C54006.

d. Terminal board arrangement shall be in accordance with Drawing 54C54396.

3.7.7.4.1 Operator's circuit. The operator's circuit shall include one Handset-Headset H-108( )/U and Cord CX-1245/U (7 feet, 6 inches), in accordance with MIL-H-26827, except there shall be no push-to-talk switch on Cord CX-1245( )/U (7 feet, 6 inches).

3.7.7.4.2 Cord circuits. Cord circuits shall terminate in Plugs PJ-051, MIL-P-642. Cords shall be of an approved type and shall be equipped with cord weights. The cords shall be color coded red, slate and green.

3.7.7.4.3 Lamps. Supervisory and recall color coded lamps shall be provided.

3.7.7.4.4 Line jacks. Line jacks for common battery lines shall be Jack JJ-042 in accordance with MIL-J-641. The line jacks for magneto lines shall be similar to Jack JJ-042 with one additional break contact. Line jacks and line signals may be strip mounted if approved components are available. Trunk jacks shall be similar to Jack JJ-042 with the addition of one "make" and one "break" contact.

3.7.7.4.5 Incoming line signals shall operate over external line loop resistances as shown below:

a. Common battery lines: 300 ohms

b. Magneto lines: 3000 ohms

c. Universal trunks: 750 ohms

The insulation resistance of all circuits shall be not less than 100 megohms.

3.7.7.4.6 Wiring. The 20 magneto and trunk lines shall be wired to one night alarm circuit. The other night alarm circuit shall be wired to the common battery lines circuit.

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3.7.7.4.7 Ground studs. Two ground studs shall be provided for terminating Electrical Lead CX-1841( )/U.

3.7.8 Telephone main frame. The telephone main frame shall consist of Main Distributing Frame, Telephone TA-270( )/GTA-6 and Main Distributing Frame, Telephone TA-285( )/GTA-6. The frames shall serve as a means for terminating and connecting outside wire lines, radio channels, various telephone units and the PBX switchboard. All openings shall be provided with gasket covers secured by approved quick detachable fasteners. It shall provide for distributing and protecting the 24-volt DC power distributed to the other telephone units. Fuse alarm circuits shall be provided in accordance with Drawing 54B54394. Cabling shall be in accordance with Drawing 51D55061.

3.7.8.1 Main Distributing Frame, Telephone TA-270( )/GTA-6. This frame shall terminate 17 each 24-pair cables and two ground wires. It shall provide 160 pairs of open space and heat coil protectors. Hand made cables shall terminate in connectors and terminal boards. Interconnecting cables shall terminate in connectors. A jack panel equipped with 140 jacks, Jack JJ-042, shall be included. A wire chief's test circuit in accordance with Drawing 54B54392 shall be included. A protector alarm circuit shall also be provided in accordance with Drawing 54A54393. Guide rings shall be provided to accomplish cross connections within the unit. One test plug and a five-position, two layer rotary type switch shall be mounted on the front panel along with two test plugs adaptable for use for Multimeter AN/PSM-6, MIL-M-4552. A test cord, terminated at one end in a plug, and at the other end in two test jacks adaptable to Multimeter AN/PSM-6 shall also be included.

3.7.8.2 Main Distributing Frame, Telephone TA-285( )/GTA-6. This frame shall be capable of terminating 48 each 26-pair cables, one five-pair cable and two ground wires. Provisions shall be made to accommodate forty 10-line operator's units. Hand made cables shall terminate in connectors and terminal boards. Interconnecting cables shall terminate in connectors. Guide rings shall be provided to accomplish cross connections within the unit. Terminal blocks shall be provided for cross connections. A fuse alarm circuit shall be provided in accordance with Drawing 54B54394.

3.7.9 Relay Rack, Telephone TA-268( )/GTA-6. All openings of the relay rack shall be provided with metal covers and gaskets secured by approved quick detachable fasteners. Terminal boards and connectors shall be provided for eight 26-pair cables, two 5-pair cables, six 24-pair cables and two ground wires. Cabling shall be in accordance with Drawing 51D55061. The rack shall be equipped

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with 60 ringdown relays, 60 ringing relays, 20 radio monitor jacks, 20 radio control toggle switches and 20 remote control relays. Fuse panels and fuse alarm circuits shall be provided. Jack panel arrangement shall be in accordance with Drawing 54C54389.

3.7.10 Power Supply PP-706( )/GTA-21. The power supply shall convert 115 volts plus or minus 10 volts, 60 cycles per second (CPS) alternating current (AC) to 24 volts direct current (DC). A transistorized frequency converter shall be furnished to supply ringing voltage. The converter shall operate from either the 115 volts AC or from the 24 volts DC. Normal operations shall be from the AC voltage with an emergency relay to switch the input to DC in case of power failure. The frequency converter output shall be 100 volts root mean square (RMS) plus or minus 10 volts, 20 CPS at 40 watts. No vibrators or other mechanical features shall be provided. The emergency relay shall not be a part of the frequency converter but shall be a separate component of the power supply with provisions for its removal and replacement as a unit. Connectors for attaching the cable from the AC supply and distributing the 24 volts DC and ringing power to the various units shall be provided. The circuit shall include an ammeter and a voltmeter. Four each Battery BB-207( )/U shall supply the 24 volts DC power. The batteries shall be charged by a hermetically sealed, oil filled selenium rectifier capable of delivering 8 amperes at normal temperature. The panel shall be wired in accordance with Drawing 51D55340. Two ground studs shall be provided for terminating Electrical Lead CX-1841( )/U.

3.7.10.1 Power panel. The power panel shall be in accordance with Drawing 54D54399. A battery compartment shall be provided and shall have ventilating ducts to openings in the side of the panel. The interior of the battery compartment shall be coated with an approved acid resistant paint. Flanges shall be provided on both sides for securely fastening to the floor. All openings shall be provided with metallic covers and gaskets secured by approved quick detachable fasteners. An instruction label for charging the batteries shall be in accordance with Drawing 54C54397.

3.7.10.2 Battery BB-207( )/U. Battery BB-207( )/U shall be in accordance with MIL-B-15072.

3.7.10.3 Electrolyte. The electrolyte for the batteries shall be a solution of sulphuric acid in accordance with O-S-801. The specific gravity of the electrolyte at a temperature of plus 26.7°C (+80°F) shall be 1.280 plus or minus 0.005.

3.7.10.4 Hydrometer. The hydrometer shall be in acoordance with GG-H-941, Class C.

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3.7.11 Control, Telephone Line C-1421( )/GTA-6. The telephone line shall contain 20 volume limiting circuits in accordance with Drawing 48D55619 and Drawing 54D54401 with the exception of the relay and toggle switches which shall appear in the relay rack. The input and output of the volume limiter circuits shall each terminate in 40 binding posts suitable for field wire.

3.7.12 Multimeter AN/PSM-6. Multimeter AN/PSM-6 shall be in accordance with MIL-M-4552.

3.7.13 Terminal Box J-618( )/GT. The terminal box shall have provisions for mounting on a pole or building. The terminal box shall be weatherproof and shall have a hinged lid with two latches. It shall contain two connecting blocks with appropriate fanning strips. The connecting blocks shall be wired to two 47 pin connectors. Distributing rings shall be provided for field wires which shall enter the box through knock-out holes. The arrangement shall be in accordance with Drawing 54D54398.

3.7.14 Telephone Circuit, Operator's TA-277( )/GTA-6. The operator's telephone circuit shall supply telephone and radio facilities for all operating positions. The unit shall contain one operator's telephone circuit (radio and telephone) and means for connecting it to 10 lines which shall have several multiple appearances in other units. Each line shall have line signals operated by relays in the relay rack and busy signals operated by keys. A grouping key shall be provided to connect the operator's circuit to adjacent units. A night alarm circuit shall be provided. The unit shall have connectors and terminal boards for one 26-pair cable and two three-conductor cords. Two jacks shall be provided for a telephone handset and handset-headset. The unit shall be as small as practicable. It shall be constructed for table-mounting and also for attaching to the frame of a plotting board with the use of an adapter. The unit shall also be capable of being mounted on a plan position indicator (PPI) of a radar set. The wiring of the telephone unit shall be in accordance with Drawing 54D54400.

3.7.15 Ground, Rod GP-26( ). The ground rod shall be a 1/2 inch diameter rod, 6 feet long, with one end cone pointed, and with a 5 inch length of copper wire attached with a clamp to the other end. One end of the copper wire shall be free for attaching the ground wire to the equipment.

3.7.16 Clamp. The clamp for the ground rod shall be galvanized steel and electro-tinned with a screw for securing the clamp to the ground rod.

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3.7.17 Adapter bracket. Stud adapter brackets shall be provided in accordance with Drawing 54B54005, 54B54006, and 54B54007.

3.7.18 Switchboard Cord CX-1220( )/U. The cord shall consist of three conductors Number 20 AWG, stranded. The cord shall be shielded and covered with a neoprene outer jacket and each conductor shall be rubber insulated. Each end shall terminate in a plug.

3.7.19 Cord Assembly, Electrical CX-2763( )/U. The cord shall be in accordance with MIL-C-10392, type WT-15/U, and shall terminate at both ends in small size connectors.

3.7.20 Electrical Lead CX-1841( )/U. The lead shall consist of one conductor, Number 8 AWG, 37 strands of tinned soft copper wire, without insulation. The lead shall terminate in a hook terminal on each end.

3.7.21 Wire WD-1( )/TT. The wire shall be in accordance with MIL-C-13294.

3.7.22 Wire, cross connecting. The wire shall be solid copper, Number 22 AWG, Thermoplastic insulation in accordance with MIL-W-16878. A 1000 foot length shall be three conductor, one pair twisted and one single. The other 1000 foot length shall be five conductor, two pair twisted and one single. The wire shall be color coded.

3.7.23 Cable Assembly, Power, Electrical CX-2795( )/U. The cable shall consist of four conductors Number 12 AWG, stranded, with natural rubber insulated conductors. Protective coverings in sequence from insulated conductors out shall be jute filler, compound filler tape and butyl compound jacket. The maximum working voltage shall be 600 volts RMS. One end shall terminate in a connector and the other end shall be sealed.

3.7.24 Cable Assembly, Telephone CX-2921( )/U. The cable shall consist of 10 conductors Number 22 AWG, stranded with natural rubber latex insulation. Protective coverings in sequence from insulated conductors out shall be jute filler, sisal paper tape and natural rubber compound. Both ends of the cable shall terminate in connectors.

3.7.25 Cable Assembly, Telephone CX-2873( )/U. The cable shall consist of 48 conductors, twisted pair Number 22 AWG, stranded wire, with natural rubber latex insulation and rubber jacket. The cable shall be terminated at each end in a connector receptacle.

3.7.26 Cable Assembly Telephone CX-2922( )/U. The cable shall consist of 52 conductors, twisted pair, Number 22 AWG stranded wire

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with natural rubber latex insulation and rubber jacket. The cable shall be terminated at each end in a connector receptacle.

3.7.27 Wire, insulated common battery hook-up. This wire shall be used for inside telephone common battery hook-up and shall consist of a twisted pair Number 22 AWG solid copper conductor, color coded red/green, rubber covered.

3.8 Weight. The maximum weight of individual components shall not exceed 400 pounds prior to packaging. (See 3.5.6)

3.9 Finish and color. Unless otherwise specified (see 6.2.e) finishes shall be in accordance with MIL-F-14072. The color of type I surfaces (as defined in MIL-F-14072) shall be color Number 15080 in accordance with FED-STD-595.

3.10 Government loaned property. When specified in the contract (see 6.2.i) the Government will loan the following to the contractor upon his request:

Item No.	Stock No.	Description	Quantity
1	5805-505-1155	Telephone Central Office Group, Manual AN/GTA-6A	1 each

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. (See 6.2.c)

4.2 Classification of tests. The inspection and testing of the telephone central group shall be classified as follows:

a. Preproduction tests.....(See 4.4)

b. Acceptance tests .....(See 4.5)

4.3 Test conditions. Unless otherwise specified, (see 4.6.4) all tests required by this specification shall be made at prevailing ambient temperature and humidity conditions.

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4.4 Preproduction testing.

4.4.1 Preproduction test samples. The preproduction test samples shall consist of models representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. (See 6.2.h)

4.4.2 Preproduction test report. When the contractor conducts the preproduction tests, he shall after completion of the tests, prepare a preproduction test report if required by the contract. This report shall conform to the requirements of MIL-STD-831. Number of copies of the report to be furnished the procuring activity shall be as specified in the contract. (See 6.2.c and 6.2.f)

4.4.3 Preproduction tests. The preproduction tests shall consist of all tests described under 4.6 "Test Methods".

4.5 Acceptance tests. The telephone central group shall be subjected to the acceptance tests. The acceptance tests shall consist of the following tests as described under 4.6 "Test methods".

a. Mechanical inspection tests .....(See 4.6.1)

b. Electrical tests .....(See 4.6.2)

c. Performance tests .....(See 4.6.3)

4.5.1 Lot. A lot shall consist of all telephone central groups submitted for acceptance at one time.

4.5.2 Sampling. For purposes of acceptance testing, sampling shall be conducted in accordance with MIL-STD-105, Inspection Level II. The acceptance number for major defects shall be zero and the AQL for minor defects shall be 2.5.

4.5.3 Classification of defects. Defects shall be classified as follows:

Category	Defects
Critical	None defined.
Major	
101	Incompleteness (such as missing parts, hardware or accessories, finishing either inadequate or

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	missing, failure to remove or neutralize solvents or flux residues that can cause harmful corrosion. Insulation sleeving missing when required. Missing locking or friction devices).
102	Difficult or improper operation of gears, adjustable and sliding parts, controls and switches. (Binding, scraping, cutting, tight spots, loose spots).
103	Evidence of leakage of oil or potting compounds from encapsulating devices.
104	Improper wire dress or cabling of wires which results in adverse electrical operation or interference with mechanical operation, leading to subsequent damage to the wire or cable due to interference with moving mechanical parts, or to short circuiting of wires due to possible abrasion or melting of insulation. Lacing of cabling too tight or too loose. Inadequate or improper anchoring of cables or wiring harnesses. Improper cable runs between components.
105	Serious conductor defects. Insulation burned, abrased, pinched which could be cause for potential short circuit. Wire sufficiently taut or loose as to incur mechanical or electrical damage. Bad surface knicks or gouges in the length of the wire. Inadequate or incorrect color coding.
106	Parts cracked, broken or otherwise defective or unsatisfactory for the purpose intended.
107	Inadequate soldering. Unsoldered joint, insufficient solder, excess solder, cold solder joint, fractured joint, resin joint.

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108	Poor fit of components in their respective positions.
109	Improper or inadequate mounting means.
110	Inadequate safety features and interlocks.
111	Components and parts not accessible for servicing.
112	Improper or inadequate bonding or grounding. Metal mesh bonding straps, or shielding on wire or cable, frayed, torn, or cut or not properly terminated or grounded.
113	Improper, inadequate or damaged threaded devices (improper type, class or size, crossed threads, heads burred or mutilated, improper thread engagement in parts, improper torque).
114	Inadequate or damaged locking or friction devices.
115	Inadequate welding. Incorrect fillet size or taper, incomplete penetration of filler metal in joint. Excessive cracks, voids, gaps in the filler or base metal. Excessive die burns. Excessive angular and thickness misalignment. Excessive warpage or dimensioned defects due to heat from the welding operation. Incomplete fusion. Undercutting. Inadequate number of spot welds to provide adequate strength. Nugget diameter insufficient.
Minor	
201	Improper, illegible, or inadequate marking or coding for identification. Improper nameplates.

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202	Finishing devects such as plating with coatings not intimately continuous or extending evenly throughout the entire finished area. Surfaces with foreign debris, not smooth or free from burns, blisters, gouge marks, checks, pits, or residuals of corrosive processing agents. Painting not of proper color or adhesion. Blisters, pits, chips, checks, cracks, flaking or unsightly scratches. Inadequate coverage, run marks, orange peel, over spray marks; dirt imbedded in coating.
203	Loose fastening and securing devices on parts or components.
204	Dimensions deviating from allowable tolerances.
205	Weight in excess of specified requirements.

4.6 Test methods.

4.6.1 Mechanical inspection tests. The telephone central group shall be given a thorough mechanical and visual inspection in accordance with the classification of defects to determine that the quality of all materials and workmanship is in compliance with the requirements of this specification.

4.6.2 Electrical tests. The telephone central group shall be given all electrical tests necessary to confirm that all circuits are inherently sound and in compliance with the requirements of this specification. The electrical tests shall include, but not necessarily be restricted to, the following.

4.6.2.1 Continuity. Each electrical component and each cord and cable shall be given a continuity test to ascertain if it is wired and connected correctly and that good electrical contact is obtained.

4.6.2.2 Operating voltages. The operating voltage at all important points shall be checked for conformance to those shown on the circuit label and schematic drawing. This shall be done with all controls set for normal operation of the equipment.

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4.6.3 Performance tests. The telephone central group shall be given all performance tests necessary to confirm that the equipment meets the performance requirements specified herein. The performance tests shall include, but not necessarily be limited to, tests of the following:

a. Signaling and talking test. A signaling and talking test shall be made on each circuit under circuit conditions simulating a completely assembled communication-monitoring system. These tests shall be made with battery voltages at 20 volts.

b. Insulation and dielectric strength test. Immediately following the test specified above, the value of the insulation resistance and dielectric strength of each circuit shall be determined between the circuit conductors and between each of the circuit conductors and the frame. A 500-volt DC Megger, or equal, shall be used in making this test.

c. D-C ripple test. The ripple of the DC output of the power panel shall not exceed 0.085 volt RMS when measured by a Ballentine voltmeter, or equal, across the DC output terminals with the battery disconnected, and the output of the rectifier adjusted to 27 volts.

d. Crosstalk test. Ten pairs of representative circuit cables shall be selected at random by the inspector from each system and tested for crosstalk. Two 600-ohm noninductive resistors, one for each circuit, shall be connected across adjacent ends of the two circuits being tested. A source of 1000-cycle potential in series with a 600-ohm noninductive load shall be connected to the open end of one circuit and vacuum-tube voltmeter. A 600-ohm noninductive shunt shall be connected to the open end of the circuit. With 2 volts RMS AC on the disturbing circuit, the vacuum-tube voltmeter reading on the disturbed circuit shall not exceed 0.001 volts.

For circuits connected to operator's units, the 1000-cycle generator shall be bridged to one circuit, and the vacuum-tube voltmeter bridged across the other circuit. The "end" of a circuit, for this purpose, shall be taken as the termination on the main frame.

4.6.4 Service conditions (environmental test). The telephone central group shall be subjected to the following environmental tests in accordance with MIL-STD-810, with exceptions as follows:

a. High temperature. Method 501, procedure I: Maintain the telephone central group at a temperature of plus 71°C (+159.8°F) for 4 hours, reduce the temperature to plus 60°C (+140°F) for 24 hours, then operate the equipment and record results.

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b. Low temperature test. Method 502, procedure I. Maintain the telephone central group at a temperature of minus 54°C (-65.2°F) for 8 hours, raise the temperature to minus 29°C (-20.2°F) for 24 hours, then operate the equipment and record results.

c. Humidity test. Method 507, procedure I. The number of continuous cycles shall be 10 for a total test time of not less than 240 hours.

d. Low pressure test. Method 500, procedure I. Maintain the telephone central group at an internal absolute pressure of 5.54 inches of mercury for 4 hours, raise the pressure to 16.8 inches of mercury for 4 hours, then operate the equipment and record results.

e. Salt fog test. Method 509, procedure I. Duration of test shall be 100 hours.

f. Fungus test. Method 508, procedure I.

g. Sand and dust test. Method 510, procedure I.

4.6.5 Service conditions (mechanical). The telephone central group shall be subjected to the following mechanical tests in accordance with MIL-T-4807:

a. Vibration test: Method 1A

b. Bounce test: Method 2A

c. Shock test: Method 3A

4.6.6 Maintainability demonstration. The maintainability of the group shall be demonstrated in accordance with Appendix A of MIL-M-26512 to a confidence level of 90 percent.

4.6.7 Production reliability testing. A production reliability test shall be performed on a quantity of production equipments in accordance with MIL-R-26474.

4.6.8 Radio frequency interference test. The telephone central group shall be tested in accordance with Class G_p of MIL-STD-826. When amount of specification readings are noted, formal testing will stop, fixes applied and retest made until the equipment meets the requirements.

4.6.9 Systems test. The telephone central group installed as in operation with a radar set, shall be given a complete systems test.

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The systems test shall be sufficiently comprehensive to insure complete compliance with the requirements specified in 3.5 and 3.6.

4.6.10 After completion of each test above the performance of the equipment shall not be below the minimum requirements specified herein.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instruction of the procuring activity (See 6.2.b).

6. NOTES

6.1 Intended use. The Telephone Central Office Group, Manual AN/GTA-6( ) covered by this specification is intended for use with fixed or mobile radar sets for intercommunication between operators and personnel of the radar set, for termination of wire and radio channels from distant points and for monitoring and supervising the work of various radio operations. The telephone central group shall also receive and control air to ground communications.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable level of preservation, packaging and packing required. (See 5.1).

c. Whether the testing will be conducted by the contractor or the procuring activity (See 4.1 and 4.4.2).

d. Finishes if other than as specified in 3.9.

e. Requirements for, and if applicable, conditions under which the production reliability and maintainability reports will be submitted by the contractor (See 3.3.10 and 3.3.11).

f. Requirements for, and if applicable, conditions under which the preproduction test report will be submitted by the contractor. (See 4.4.2)

g. Requirements for, and if applicable, conditions under which nomenclature data will be submitted by the contractor. (See 3.4)

h. The quantity of preproduction test samples. (See 4.4.1)

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i. Conditions under which Government loaned property will be made available to the contractor (See 6.10).

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Air Force - 71

REVIEWER:
Air Force - 71, 11

PREPARING ACTIVITY:
Air Force - 71

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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